

Povketya NHOR

Data Engineering · Machine Learning · AI

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PROFESSIONAL SUMMARY

Junior data professional with experience in machine learning, data analysis, data visualization, data engineering, and AI-focused data preparation. Experienced with Python, SQL, Power BI, Scikit-learn, TensorFlow, SHAP, ETL workflows, dashboards, and database technologies. Interested in building practical data and AI solutions that support better decisions and solve real-world problems.

EXPERIENCE

Teaching Assistant — Data Science and Programming Apr 2025 — Apr 2026

Royal University of Phnom Penh

- Assisted with Python, SQL, programming fundamentals, and introductory data science for classes serving more than 50 students.
- Supported laboratory exercises, assignments, debugging, and projects with practical guidance.
- Explained data-processing and analytical concepts through applied examples and mentoring.

Data Engineer Mar 2025 — Jul 2025

COPAI AI Innovation Lab

- Collected and processed Khmer-language data through web scraping and multi-source dataset curation.
- Cleaned, standardized, annotated, validated, and quality-checked structured and unstructured text data.
- Helped build a unified knowledge base for RAG and Khmer AI chatbot applications.
- Prepared training datasets for deployment on MISTI's High-Performance Computing platform.
- Collaborated with researchers and AI engineers on machine-learning and NLP solutions.

SELECTED PROJECTS

Explainable AI for Diabetes Risk Prediction and Actionable Insights

Prepared and aligned BRFSS and Cambodia STEPS data; compared classical and gradient-boosting classifiers; addressed class imbalance and threshold selection; used SHAP for explanations; and built a Streamlit risk-communication application.

Python · Scikit-learn · XGBoost · LightGBM · CatBoost · SHAP · Streamlit

RFM Customer Segmentation Analysis

Analyzed 4,906 transactions from 995 customers and created 11 RFM-based customer groups with retention, re-engagement, and sales recommendations.

Python · Pandas · Matplotlib · Customer Analytics

HomeBuild Retail Sales and Digital Transformation Analysis

Built Power BI dashboards and DAX measures across 285 transactions and 20 customers; identified 30.88% Bakong payment adoption and developed digital-transformation recommendations.

Power BI · DAX · Excel · OneDrive

Loan Risk Prediction System

Developed a TensorFlow prediction workflow and integrated it into a role-based Django and PostgreSQL application for administrators and customers.

Python · TensorFlow · Django · PostgreSQL

Facial Recognition for Secure Authentication in the Quantum Era

Created preprocessing and inference workflows and integrated facial recognition into a Django application. First Prize, 7th Engineering Day Awards 2024.

Python · TensorFlow · Django · PostgreSQL

Personal Financial Analyst

Built a personal-finance application with role-based access, authentication, importing, CRUD workflows, and dashboards for income, expenses, debt, and savings goals.

Django · PostgreSQL · ReactJS

Additional Projects

Diamond Shop Management System · Performance Tracker Dashboard · PETER TECH Web Store. Full details and screenshots are available on the portfolio.

EDUCATION

Bachelor’s Degree in Data Science and Engineering	2023 – 2026
Royal University of Phnom Penh	

ACHIEVEMENT

First Prize
Facial Recognition for Secure Authentication in the Quantum Era
7th Engineering Day Awards 2024 · Royal University of Phnom Penh

TECHNICAL SKILLS

Programming	Python, SQL
Analysis	Pandas, NumPy, Matplotlib, data cleaning, EDA, feature engineering, statistical analysis
ML & AI	Scikit-learn, TensorFlow, XGBoost, LightGBM, CatBoost, SHAP, classification, model evaluation, explainable AI
Business Intelligence	Power BI, DAX, dashboard design, KPI reporting, Excel, Superset
Data Engineering	ETL, Apache Airflow, Apache NiFi, Oracle, PostgreSQL, MySQL, data warehousing, dimensional modeling, data quality
Development	Streamlit, Django, ReactJS, FastAPI, Next.js, Docker, Git, GitHub

TRAINING

- ASEAN Data Science Explorers 2024 — SAP Analytics Cloud & SAP Build Apps

LANGUAGES

- Khmer — Native
- English — Intermediate